BX-221 N-Channel Enhancement Mode Power MOSFET, 80 V, 18 A

Features

- V_{DS} = 80 V, I_D = 18 A
- $R_{DS(on)}$ < 27 mΩ @ V_{GS} = 10 V
- Low gate charge (typ. 17 nC)
- 100% avalanche tested
- RoHS compliant, halogen free

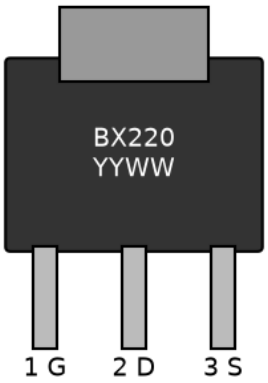
Applications

- DC/DC converters
- Motor drives
- Load switching

Description

The BX-221 uses an advanced trench process to provide excellent $R_{DS(on)}$ with low gate charge. It is suitable for high-frequency switching and synchronous rectification. The device is offered in a TO-252 (DPAK) package and is qualified according to AEC-Q101.

TO-252 (DPAK) - Top View



Tab = Drain | Marking code: B22

Absolute Maximum Ratings (T_A = 25 °C unless otherwise noted)

Parameter	Symbol	Value	Unit
Drain-Source Voltage	VDS	80	V
Gate-Source Voltage	VGS	±20	V
Continuous Drain Current, TC = 25 °C	ID	18	A
Continuous Drain Current, TC = 100 °C	ID	12.7	A
Pulsed Drain Current (1)	IDM	72	A
Single Pulse Avalanche Energy (2)	EAS	61	mJ
Power Dissipation, TC = 25 °C	PD	52	W
Operating Junction and Storage Temperature	TJ, TSTG	-55 to +175	°C

Electrical Characteristics (T_J = 25 °C unless otherwise noted)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Static Characteristics						
Drain-Source Breakdown Voltage	V(BR)DSS	VGS = 0 V, ID = 250 µA	80	-	-	V
Zero Gate Voltage Drain Current	IDSS	VDS = 80 V, VGS = 0 V	-	-	1	µA
Gate-Body Leakage Current	IGSS	VGS = ±20 V, VDS = 0 V	-	-	±100	nA
Gate Threshold Voltage	VGS(th)	VDS = VGS, ID = 250 µA	1.4	2.0	2.7	V
Static Drain-Source On-Resistance	RDS(on)	VGS = 10 V, ID = 11 A	-	21	27	mOhm
		VGS = 4.5 V, ID = 8 A	-	29	37	mOhm
Forward Transconductance	gFS	VDS = 5 V, ID = 11 A	-	24	-	S
Dynamic Characteristics						
Input Capacitance	Ciss	VDS = 30 V, VGS = 0 V,	-	1320	-	pF
Output Capacitance	Coss	f = 1 MHz	-	142	-	pF
Reverse Transfer Capacitance	Crss		-	41	-	pF
Total Gate Charge	Qg	VDS = 30 V, ID = 11 A,	-	17	25	nC
Gate-Source Charge	Qgs	VGS = 10 V	-	4.6	-	nC
Gate-Drain Charge	Qgd		-	5.3	-	nC
Switching Characteristics						
Turn-On Delay Time	td(on)	VDD = 30 V, ID = 11 A,	-	8	-	ns

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Rise Time	tr	VGS = 10 V, RG = 3.3 Ohm	-	14	-	ns
Turn-Off Delay Time	td(off)		-	27	-	ns
Fall Time	tf		-	8	-	ns
Drain-Source Diode Characteristics						
Diode Forward Voltage	VSD	VGS = 0 V, IS = 11 A	-	0.86	1.2	V
Reverse Recovery Time	trr	IF = 11 A, di/dt = 100 A/µs	-	31	-	ns
Reverse Recovery Charge	Qrr		-	26	-	nC

Notes: (1) Repetitive rating; pulse width limited by maximum junction temperature. (2) Starting TJ = 25 °C, L = 0.5 mH, IAS = 14.7 A.

Typical Characteristics

Fig. 1 Output Characteristics

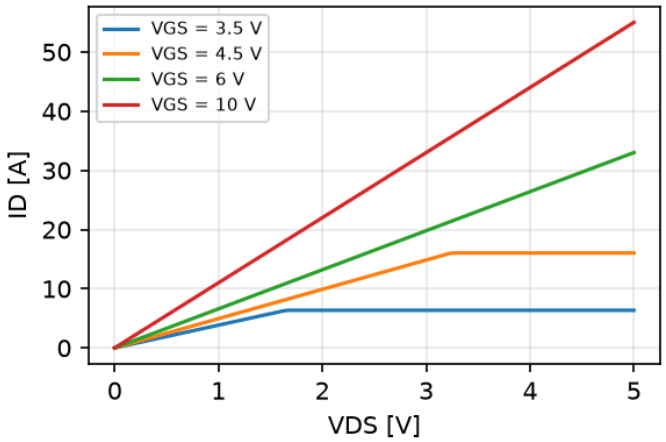
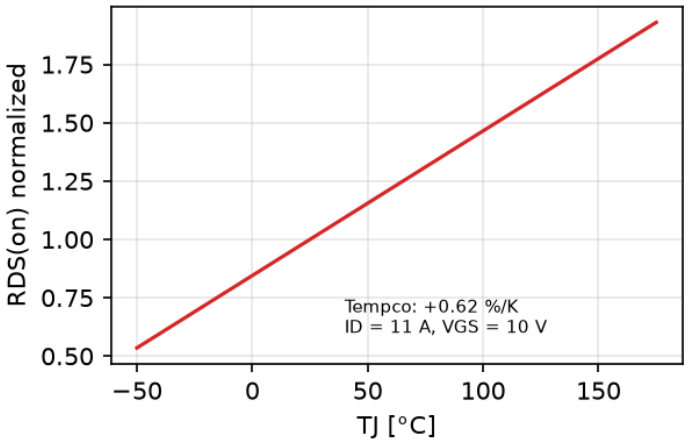


Fig. 2 On-Resistance vs. Temperature



Ordering Information

Part Number	Package	Marking	Packing	MOQ
BX-221-DPK-T	TO-252	B21	Tape & Reel, 2500 pcs	2500
BX-221-DPK-B	TO-252	B21	Tube, 75 pcs	750